

The complete Crouzeix theorem: optimal similarity and a common positive boundary representation

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September 23, 2026

Abstract

We resolve the complete Crouzeix conjecture by proving the sharp constant-two numerical-range inequality for every bounded operator on a complex Hilbert space and every matrix-valued polynomial. No separability assumption is needed. The closure of the numerical range is a complete 2-spectral set, and the bound extends to finite matrix-valued functions holomorphic near that closure. For a finite matrix and a bounded convex domain containing its numerical range with regular real-analytic Jordan boundary, the optimal similarity making its conformal disk image contractive is attained with condition number at most two. For the similar matrix, one continuous positive boundary density of mass the identity represents the evaluation of every matrix-valued function holomorphic near the closed domain.

1 The structural question

Can one choose a single controlled change of inner product and a single positive boundary density to represent all matrix-valued analytic evaluations of a matrix? We answer this question on an analytic convex neighborhood of its numerical range. The metric and density depend on the matrix and the enclosing domain, but neither depends on the coefficient size or the function being evaluated. The resulting complete inequality also holds for every bounded operator on an arbitrary complex Hilbert space, by a finite compression argument in Section 6.

Coordinates and evaluation

Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and $\mathbb{T} = \partial\mathbb{D}$, with normalized angular measure $d\sigma(e^{i\theta}) = d\theta/(2\pi)$. For $A \in M_n(\mathbb{C})$ its numerical range is

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, x^*x = 1\}.$$

All operator norms are Hilbert-space operator norms, and tensor products are Hilbert tensor products with the base space first. For a matrix-valued function v holomorphic near the spectrum, $v[A]$ denotes holomorphic evaluation: on a Jordan block $J = \beta I + N$ of size s it is $\sum_{j=0}^{s-1} N^j \otimes v^{(j)}(\beta)/j!$, followed by the change of basis on the first factor. In particular, $v[A] = \sum_j A^j \otimes B_j$ when $v(z) = \sum_j B_j z^j$. Inner products are conjugate-linear in the first argument, and $U \preceq V$ means that $V - U$ is positive semidefinite. The symbol $*$ denotes the adjoint on a base or coefficient Hilbert space, $^\top$ ordinary transpose, and † the adjoint of an operator on a function space.

Call Ω *admissible* when it is a bounded open convex subset of $\mathbb{C}$ with regular real-analytic Jordan boundary.

Lemma 1.1 (Conformal coordinates). *For every admissible Ω there is a conformal bijection $f : \Omega \rightarrow \mathbb{D}$ such that f and f^{-1} extend univalently through $\overline{\Omega}$ and $\overline{\mathbb{D}}$, respectively. There is also a conformal bijection $G : \{|t| > 1\} \rightarrow \mathbb{C} \setminus \overline{\Omega}$ that extends univalently to $\{|t| > r\} \cup \{\infty\}$ for some $0 < r < 1$, as a sphere map, with a simple pole at infinity and $G(\mathbb{T}) = \partial\Omega$. On $\mathbb{T}$, $tG'(t)$ is an outward normal and $\psi = f \circ G$ is an orientation-preserving smooth circle diffeomorphism.*

The complete local Riemann-mapping, harmonic-reflection, and collar argument is in Appendix A.

Fix such an Ω with $W(A) \subset \Omega$, and choose its coordinates f, G from Lemma 1.1. Put $T = f(A)$ and define

$$\kappa^2 = \min\{\tau \in \mathbb{R} : I \preceq H \preceq \tau I, T^*HT \preceq H \text{ for some } H = H^*\}. \quad (1.1)$$

A strictly feasible pair means that $H - I$, $\tau I - H$, and $H - T^*HT$ are all positive definite. Existence and attainment of the minimum are part of the first conclusion.

Theorem 1.2 (The optimal metric). *For every admissible Ω , every finite $n \geq 1$, every $A \in M_n(\mathbb{C})$ with $W(A) \subset \Omega$, and the interior conformal coordinate f above, the problem (1.1) is strictly feasible and has an attained minimum $1 \leq \kappa^2 \leq 4$. If H is a minimizing metric, then*

$$S = H^{1/2}, \quad A' = SAS^{-1}, \quad D = Sf(A)S^{-1} = f(A')$$

satisfy

$$D^*D \preceq I, \quad \rho(D) < 1, \quad \|S\|\|S^{-1}\| = \kappa \leq 2.$$

Theorem 1.3 (A common positive representation). *With the same Ω, A, f, G and any chosen minimizing metric in Theorem 1.2, there is one continuous function $\Lambda : \mathbb{T} \rightarrow M_n(\mathbb{C})$ satisfying*

$$\Lambda(t) \succeq 0, \quad \int_{\mathbb{T}} \Lambda(t) d\sigma(t) = I_n,$$

such that, for every positive integer m and every $M_m(\mathbb{C})$ -valued function v holomorphic on a neighborhood of $\overline{\Omega}$,

$$v[A'] = \int_{\mathbb{T}} \Lambda(t) \otimes v(G(t)) d\sigma(t), \quad \|v[A]\| \leq \kappa \max_{\overline{\Omega}} \|v\|. \quad (1.2)$$

The same S , Λ , and κ work for all m and v .

The complete inequality and previous work

For a bounded operator A on a nonzero complex Hilbert space $\mathcal{H}$, write $W(A) = \{\langle x, Ax \rangle : \|x\| = 1\}$. For a matrix polynomial $P(z) = \sum_{j=0}^d B_j z^j$, with $B_j \in M_m(\mathbb{C})$, put $P[A] = \sum_{j=0}^d A^j \otimes B_j$ on $\mathcal{H} \otimes \mathbb{C}^m$. The operator formulation of the complete Crouzeix conjecture asks whether $\|P[A]\| \leq 2 \sup_{W(A)} \|P\|$ holds independently of the Hilbert space, the finite coefficient size m , and the degree d . The scalar problem is the case $m = 1$. Theorems 1.2 and 1.3, together with analytic convex outer approximation, give the finite-matrix inequality in Corollary 6.1. Corollary 6.3 transfers it to all bounded Hilbert-space operators, including those on nonseparable spaces, and gives the corresponding rational and holomorphic bounds on $\overline{W(A)}$. The optimal similarity and the common density are the finite-dimensional structural conclusions proved before these passages.

Crouzeix formulated the scalar constant-two question in 2004 and obtained a universal complete bound of 11.08 in 2007 [9, 10]. Delyon and Delyon's positive integral representations form an

important part of the numerical-range method [12]. Crouzeix and Palencia used a positive double-layer potential and a conjugate-data Cauchy transform to obtain the complete bound $1 + \sqrt{2}$ [11]; Ransford and Schwenninger simplified its final norm argument [21]. Recent preprints by Jin, Lorist and Schwenninger, and Luo present scalar constant-two proofs [14, 16, 17]. Lorist and Schwenninger’s Theorem 3 covers bounded Hilbert-space operators; their Section 3(iv) explains why its commutation argument does not immediately pass to matrix coefficients. Åhag, Czyż, and Virtanen prove the complete constant-two estimate for base matrices of order at most three, with arbitrary coefficient size [1, Theorem 1.1]. The present comparison retains both matrix multiplication orders in all finite base dimensions, which suffices for the bounded-operator consequence by compression.

The abstract connection with controlled similarity is classical. Paulsen’s similarity theorem identifies the optimal condition number with the completely bounded norm [19, Section 2, unnumbered theorem]; Arveson’s dilation theorem relates complete contractivity to a normal boundary dilation [3, Theorem 1.2.2]. For numerical-range ellipses in base dimension two, Badea, Crouzeix, and Delyon construct a conformal similarity with condition number at most two [4, Theorem 5.2 and Corollary 5.3]. Here we construct the minimizing metric, its endpoint extremal data, and the continuous density in a fixed boundary parameter, and prove $\kappa \leq 2$ for the optimal metric constant in (1.1).

The direct proof in [18] reaches the polynomial inequality from the singular vectors of the polynomial being evaluated. Its exterior kernel and original-matrix resolvent calculations also occur here, with complete local proofs. The present argument instead chooses one optimal metric first and proves a representation valid for every coefficient size and every analytic test function. The two manuscripts have separate complete proof paths.

How the two conclusions are proved

Section 2 first attains the metric minimum and constructs its positive disk density before estimating κ . The boundary change of coordinates

$$t \in \mathbb{T} \xrightarrow{G} G(t) \in \partial\Omega \xrightarrow{f} \psi(t) \in \mathbb{T}$$

has a positive angular Jacobian. Including it in the density gives Λ . Its mass-one identity makes the map $b \mapsto [t \mapsto (\Lambda(t)^{1/2} \otimes I_m)b]$ an isometric embedding of $\mathbb{C}^n \otimes \mathbb{C}^m$ into boundary L^2 functions. Multiplying there by $I_n \otimes v(G(t))$ and applying the embedding’s adjoint gives $v[A']$. This is the common compression at every coefficient size. Figure 1 displays the two boundary parameters.

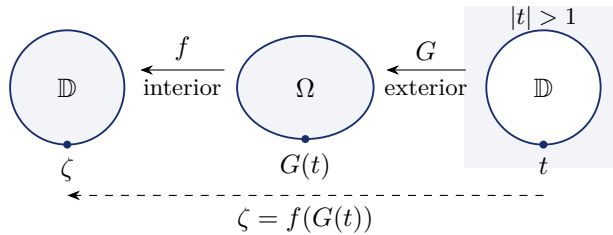


Figure 1: The two conformal coordinates, shown schematically; Ω need not be an ellipse. The Cauchy projection uses the exterior boundary parameter t and its uniform circle measure. The contraction representation uses the interior disk coordinate ζ ; passing between them requires the positive Jacobian of the boundary map $f \circ G$. The shaded exterior on the right is the domain of G .

If $\kappa > 1$, finite-dimensional separation supplies unit vectors $x, y \in \mathbb{C}^n \otimes \mathbb{C}^n$ in the κ and 1 eigenspaces of $S \otimes I_n$, respectively. A unitary realization gives a contractive $M_n(\mathbb{C})$ -valued disk function $F_{\mathbb{D}}$ with $F_{\mathbb{D}}[D]x = y$. This auxiliary function has coefficient size n ; its role is to test one base-space metric, not to restrict the independently chosen target size m .

Once the extremal vectors are available, Section 3 introduces the local boundary tools needed to compare them: the exterior kernel, the actual Cauchy adjoint, both analytic multiplication orders, and the positive resolvent of the original A . In Section 4, equality in the common compression gives the forward and adjoint relations between the boundary vectors $x_{\Lambda}(t) = (\Lambda(t)^{1/2} \otimes I_n)x$ and $y_{\Lambda}(t) = (\Lambda(t)^{1/2} \otimes I_n)y$. Partial traces of their outer products sum the diagonal base-space blocks, leaving coefficient-space matrix fields. An ordinary transpose fixed by the tensor convention identifies their means, including an entirely vanishing matrix cross mean. The original- A resolvent produces a second positive block. Analytic and adjointed cross tests relate the two blocks with factors κ and κ^{-1} , respectively.

The decisive comparison is between the cross density and the two diagonal densities in this second positive block. We use boundary L^2 norms formed with the Hilbert–Schmidt matrix norm. Positivity bounds twice the squared norm of the cross density by the sum of the squared norms of the diagonal densities. Both diagonal norms are controlled by the same nonzero projected cross density from the first system. The zero matrix mean separates the two projected cross terms into orthogonal Fourier subspaces, so the factor κ gives a lower bound for the second cross density. Section 5 combines these bounds to prove $\kappa \leq 2$. This completes the metric theorem and returns to the common density at every target size. Section 6 applies the result to polynomials and passes to arbitrary finite-matrix numerical ranges, including points and segments. It then uses finite compressions and holomorphic approximation to obtain the complete inequality for bounded operators on arbitrary complex Hilbert spaces. Appendix A provides the full local conformal construction used throughout.

2 The metric, its representation, and its extremal data

2.1 A metric for the disk coordinate

The evaluation rules used below follow directly from the finite Jordan Taylor formula. Its product rule preserves the order of matrix coefficients; conjugating the Jordan form proves similarity on the base factor. For scalar composition, substitute the nilpotent matrix $f(J) - f(\beta)I$ into the Taylor series of the outer function at $f(\beta)$. The finite Taylor composition identity gives $(v \circ f)[J] = v[f(J)]$, and the same formula gives $\text{spec}(f(A)) = f(\text{spec}(A))$. No diagonalizability is assumed. The Jordan resolvent expansion and Cauchy’s derivative formula give the equivalent contour-integral evaluation.

Since $\text{spec}(A) \subset W(A) \subset \Omega$, spectral mapping gives $\rho(T) < 1$. The positive-metric feasibility criterion is classical [24, Theorem 1]; we construct the strictly feasible witness. Choose $\rho(T) < \eta < 1$. Jordan normal form supplies C_{η} with $\|T^j\| \leq C_{\eta}\eta^j$. Hence

$$H_0 = \sum_{j \geq 0} T^{*j} T^j$$

converges in norm, $H_0 \succeq I$, and $H_0 - T^* H_0 T = I$. For $c > 1$ and $\tau > \|cH_0\|$, the pair (τ, cH_0) has all three matrix slacks in (1.1) positive definite. This is strict feasibility of the optimization problem. Feasible pairs below any fixed feasible upper bound form a closed bounded subset of a finite-dimensional space, so a minimum is attained. Every feasible τ is at least one.

Fix a minimizing H and put $S = H^{1/2}$. Congruence gives $D^*D = S^{-1}T^*HTS^{-1} \preceq I$; similarity gives $\rho(D) < 1$. The inequalities $I \preceq H \preceq \kappa^2 I$ give $\|S\|\|S^{-1}\| \leq \kappa$. The optimizer itself may have zero slacks. In particular $\|D\| = 1$ is allowed.

The minimum also measures the best condition number of a similarity to a contraction. If R is invertible and $\|RTR^{-1}\| \leq 1$, then $H_R = R^*R$ satisfies $T^*H_RT \preceq H_R$. Dividing H_R by its least eigenvalue gives a feasible metric with $\tau = \|R\|^2\|R^{-1}\|^2$. Consequently $\kappa \leq \|R\|\|R^{-1}\|$. Applying this to the minimizing $S = H^{1/2}$, for which the reverse inequality was just proved, gives

$$\kappa = \min\{\|R\|\|R^{-1}\| : R \text{ invertible, } \|RTR^{-1}\| \leq 1\}. \quad (2.1)$$

The minimum is attained by the positive matrix S , with $\|S\|\|S^{-1}\| = \kappa$. This identity holds for any fixed finite T with an attained feasible metric problem; the stability assumption above supplies such feasibility. The bound $\kappa \leq 2$ proved below uses the additional conformal and numerical-range hypotheses.

2.2 One density for every coefficient size

Proposition 2.1 (One density for every coefficient size). *For the fixed minimizing metric, the representation (1.2) holds, before any bound on κ is proved, with*

$$\Phi_D(\zeta) = (I - D/\zeta)^{-1} + [(I - D/\zeta)^{-1}]^* - I, \quad (2.2)$$

$$j(t) = \frac{tG'(t)f'(G(t))}{\psi(t)} > 0, \quad \Lambda(t) = j(t)\Phi_D(\psi(t)). \quad (2.3)$$

The disk-resolvent formula is the positive representation of Delyon and Delyon [12, Section 2, equation (7)]; the common compression also belongs to the dilation tradition [25, Theorem I]. We express the density in the exterior parameter and prove every coefficient-size claim.

Proof. For $|\zeta| = 1$, $E_\zeta = I - D/\zeta$ is invertible and

$$E_\zeta^* \Phi_D(\zeta) E_\zeta = I - D^*D \succeq 0.$$

Thus Φ_D is positive and continuous. The strict spectral-radius bound, rather than a strict norm bound, implies $\sum_{j \geq 0} \|D^j\| < \infty$. Consequently the following series is absolutely uniformly convergent on the circle:

$$\Phi_D(\zeta) = I + \sum_{j \geq 1} (\zeta^{-j} D^j + \zeta^j D^{*j}), \quad \int_{\mathbb{T}} \zeta^j \Phi_D(\zeta) d\sigma(\zeta) = D^j \quad (j \geq 0).$$

For any m and any M_m -valued h holomorphic near $\overline{\mathbb{D}}$, integration of its uniformly convergent Taylor series gives

$$h[D] = \int_{\mathbb{T}} \Phi_D(\zeta) \otimes h(\zeta) d\sigma(\zeta).$$

Apply this with $h = v \circ f^{-1}$. The conformal collar makes h holomorphic near $\overline{\mathbb{D}}$, and scalar composition gives $h[D] = v[A']$. Locally writing $\psi(e^{i\theta}) = e^{in(\theta)}$ shows that $j(e^{i\theta}) = \eta'(\theta) > 0$. Changing variables $\zeta = \psi(t)$ proves the first identity in (1.2). The zeroth moment gives $\int \Lambda d\sigma = I$.

For each m , define

$$V_m : \mathbb{C}^n \otimes \mathbb{C}^m \longrightarrow L^2(\mathbb{T}, d\sigma; \mathbb{C}^n \otimes \mathbb{C}^m), \quad (V_m b)(t) = (\Lambda(t)^{1/2} \otimes I_m) b.$$

Normalization makes V_m an isometry. If M_v multiplies by $I_n \otimes v(G(t))$, then

$$v[A'] = V_m^\dagger M_v V_m, \quad \|v[A']\| \leq \max_{\Omega} \|v\|.$$

Similarity on the first factor gives the remaining estimate with constant κ . Neither the definition of S nor that of Λ depends on m or v . $\square$

2.3 Endpoint separation and an exact extremal

To bound the metric, we need equality for one auxiliary matrix function. The construction passes from real separation to a dual balance, then to endpoint supports, and finally to a finite unitary realization. Use the vectorization convention

$$\text{vec}(X) = \sum_{j=1}^n X e_j \otimes e_j, \quad \|\text{vec}(X)\|^2 = \text{tr}(X X^*).$$

Proposition 2.2 (Endpoint extremal). *If $\kappa > 1$, there are $X, Y \in M_n(\mathbb{C})$ with*

$$\text{tr}(X X^*) = \text{tr}(Y Y^*) = 1, \quad S X = \kappa X, \quad S Y = Y, \quad X^* Y = 0,$$

and an M_n -valued rational function $F_{\mathbb{D}}$, holomorphic on a neighborhood of $\overline{\mathbb{D}}$ and contractive on $\overline{\mathbb{D}}$, such that, for $x = \text{vec}(X)$ and $y = \text{vec}(Y)$,

$$F_{\mathbb{D}}[D]x = y.$$

In particular $F_{\Omega} = F_{\mathbb{D}} \circ f$ is holomorphic near $\overline{\Omega}$ and satisfies $F_{\Omega}[A']x = y$. Its boundary trace $F = F_{\Omega} \circ G$ is contractive.

Proof. We first obtain positive matrices X_0 and Y_0 supported on the κ^2 and 1 eigenspaces of H , respectively. A third positive matrix Z_0 will express their difference as $Z_0 - T Z_0 T^*$; this balance will provide the equal Gram matrices needed for the unitary realization.

Separation, including a possible contact point. In the real vector space $\mathbb{R} \oplus \text{Herm}_n^3$, the affine set

$$\mathcal{A}_T = \{(s, J - I, sI - J, J - T^* J T) : s \in \mathbb{R}, J = J^*\}$$

is disjoint from the open convex set

$$\mathcal{O} = \{(s, U, V, W) : s < \kappa^2, U, V, W \succ 0\}.$$

Here is the finite-dimensional separation argument used for these sets; the general convex-separation framework is treated in [22, Section 11, Theorem 11.2]. Write $\mathcal{A}_T = a_0 + L$, and let π project orthogonally onto $L^\perp$. The set $\mathcal{K} = \pi(\mathcal{O} - a_0)$ is nonempty, open in $L^\perp$, and convex, and it does not contain zero. If $0 \notin \overline{\mathcal{K}}$, a nearest point z_0 in the closed convex set $\overline{\mathcal{K}}$ satisfies $\langle z_0, c \rangle \geq \|z_0\|^2 > 0$ for every $c \in \mathcal{K}$.

If $0 \in \overline{\mathcal{K}}$, fix $c_* \in \mathcal{K}$ and set $z_j = -c_*/j$. Each z_j is outside $\overline{\mathcal{K}}$: otherwise zero would lie on the open segment joining an interior point c_* to a point of the closure, and hence would belong to $\mathcal{K}$. To verify that segment fact, take a ball about c_* inside $\mathcal{K}$, approximate the other endpoint by points of $\mathcal{K}$, and use convexity; a ball about each fixed interior point of the limiting segment is still contained in $\mathcal{K}$. Let y_j be nearest to z_j in $\overline{\mathcal{K}}$. Since zero belongs to this closed set, $\|y_j - z_j\| \leq \|z_j\|$, so $y_j \rightarrow 0$. A subsequence of $\nu_j = (y_j - z_j)/\|y_j - z_j\|$ tends to a unit vector ν . The nearest-point

inequality gives $\langle \nu_j, c - y_j \rangle \geq 0$, hence $\langle \nu, c \rangle \geq 0$ for every $c \in \mathcal{K}$. In the first case use $\nu = z_0$. In either case $\ell(v) = -\langle \nu, \pi v \rangle$ is nonzero and $\ell(a) \geq \ell(o)$ for all $a \in \mathcal{A}_T$, $o \in \mathcal{O}$.

Because $\mathcal{A}_T$ is affine, ℓ is constant on it. Boundedness above on $\mathcal{O}$ forces the signs

$$\ell(s, U, V, W) = \lambda s - \text{tr}(Y_0 U) - \text{tr}(X_0 V) - \text{tr}(Z_0 W), \quad \lambda \geq 0, \quad X_0, Y_0, Z_0 \succeq 0.$$

If $\lambda = 0$, strict feasibility produces a negative value on $\mathcal{A}_T$, while separation requires a value at least $\sup_{\mathcal{O}} \ell = 0$. Thus $\lambda > 0$; scale it to one. Constancy on $\mathcal{A}_T$ then gives

$$X_0 - Y_0 = Z_0 - T Z_0 T^*, \quad \text{tr } X_0 = 1. \quad (2.4)$$

The constant value is $\text{tr } Y_0 \geq \kappa^2$. At (κ^2, H) it is

$$\kappa^2 - \text{tr}(Y_0(H - I)) - \text{tr}(X_0(\kappa^2 I - H)) - \text{tr}(Z_0(H - T^* H T)) \leq \kappa^2.$$

Each subtracted trace is nonnegative, so all vanish and $\text{tr } Y_0 = \kappa^2$.

Endpoint support and balance. Write $\|B\|_{\text{HS}} = (\text{tr}(B^* B))^{1/2}$ for the unnormalized Hilbert–Schmidt norm. For positive matrices B, C , $\text{tr}(BC) = \|C^{1/2} B^{1/2}\|_{\text{HS}}^2$; vanishing implies that the range of B lies in the kernel of C . Therefore X_0 is supported on the κ^2 eigenspace of H , and Y_0 on its 1 eigenspace.

Choose square factors and a positive matrix Q by

$$X X^* = \kappa^{-2} S X_0 S, \quad Y Y^* = \kappa^{-2} S Y_0 S, \quad Q = \kappa^{-2} S Z_0 S.$$

The support and trace relations give the asserted norms and endpoint identities. Their column ranges are orthogonal because $\kappa > 1$, so $X^* Y = 0$ as a matrix. Congruence of (2.4) gives

$$X X^* - Y Y^* = Q - D Q D^*. \quad (2.5)$$

Unitary realization and transpose. The balance-to-isometry step is the classical interpolation mechanism of [15, Section 3 of the versioned preprint]; its particular endpoint data and finite-dimensional realization are established here. Put $L = Q^{1/2}$. The rows $R_1 = [L \ Y]$ and $R_2 = [DL \ X]$ have equal products with their adjoints. The map $R_2^* v \mapsto R_1^* v$ is therefore a well-defined isometry of their ranges. Extend it to a unitary using equal-dimensional orthogonal complements, then take its adjoint. This yields

$$[L \ Y] = [DL \ X] \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ unitary.}$$

Thus $L = DLa + Xc$ and $Y = DLb + Xd$. Since $\|a\| \leq 1$ and $\rho(D) < 1$, iteration gives the norm-convergent identity

$$Y = Xd + \sum_{j \geq 0} D^{j+1} X c a^j b. \quad (2.6)$$

The transfer-function norm identity is the unitary-colligation identity of [8, Section 3.1, equations (3.1)–(3.4)]. Define, for $|\zeta| < 1$,

$$R(\zeta) = d + \zeta c(I - \zeta a)^{-1} b, \quad F_{\mathbb{D}}(\zeta) = R(\zeta)^{\top}.$$

For $e \in \mathbb{C}^n$ and $v = (I - \zeta a)^{-1} b e$, unitarity gives

$$U \begin{pmatrix} \zeta v \\ e \end{pmatrix} = \begin{pmatrix} v \\ R(\zeta) e \end{pmatrix}, \quad \|e\|^2 - \|R(\zeta) e\|^2 = (1 - |\zeta|^2) \|v\|^2 \geq 0.$$

Hence R is contractive. Ordinary transpose preserves its norm and holomorphy. Its rational entries are bounded in $\mathbb{D}$, so every apparent pole on $\mathbb{T}$ is removable: a genuine pole would diverge on an interior radial approach. After cancellation, finitely many remaining poles stay away from $\overline{\mathbb{D}}$. This supplies a common neighborhood of that closed disk on which $F_{\mathbb{D}}$ is holomorphic; continuity retains contractivity on the circle.

Finally,

$$(D^k \otimes B) \operatorname{vec}(X) = \operatorname{vec}(D^k X B^{\top}).$$

The Taylor coefficients of $F_{\mathbb{D}}$ are the transposes of d and $ca^j b$. Substitution in (2.6) proves $F_{\mathbb{D}}[D]x = y$. The conformal collar and scalar composition give the claims for F_{Ω} . Rationality is asserted in the disk variable; F_{Ω} is only required to be holomorphic near $\overline{\Omega}$. $\square$

3 Local boundary tools

We now prepare the comparison that will bound the optimal metric. The exterior coordinate supplies a positive kernel with two normalized marginals. Its Fourier action controls a Cauchy projection, while the original numerical range supplies a second positive density. We prove the shared exterior and resolvent calculations here. They also form the analytic part of the direct polynomial proof [18, Sections 2–3]; each required argument is included in the present paper. A reader who has checked [18, Theorems 2.1 and 3.1] may compare the statement of Proposition 3.1 using the notation in Remark 3.2, and then resume at the physical Cauchy integral preceding Proposition 3.3. The projection proof and its ordered applications remain part of the present argument.

Proposition 3.1 (Exterior coefficients and the original-matrix resolvent). *Assume the admissible Ω and the exterior map G supplied above. Write $G(t) = at + b + O(t^{-1})$ at infinity, where $a \neq 0$, and $q_G(t) = tG'(t)$. There are scalar polynomials b_k , with $b_0 = 1$, $\deg b_k = k$, and leading coefficient a^{-k} , defined by*

$$\frac{q_G(t)}{G(t) - z} = \sum_{k \geq 0} b_k(z) t^{-k} \quad \text{at infinity.} \quad (3.1)$$

There is also the mixed expansion

$$B(w, t) = \frac{q_G(t)}{G(t) - G(w)} - \frac{t}{t - w} = \sum_{i, j \geq 1} \beta_{ij} w^{-i} t^{-j}, \quad b_j(G(w)) = w^j + \sum_{i \geq 1} \beta_{ij} w^{-i} \quad (j \geq 1). \quad (3.2)$$

Here B is holomorphic through the diagonal and both infinities of the exterior product collar. For each $r < \rho < 1$ there is a constant C_ρ such that $|\beta_{ij}| \leq C_\rho \rho^{i+j}$; the series converges normally on the product collar and absolutely uniformly on $\mathbb{T}^2$. The kernel

$$m(w, t) = 1 + B(w, t) + \overline{B(w, t)}$$

is continuous and nonnegative, with each normalized-circle marginal one.

For every finite n and every original matrix $A \in M_n(\mathbb{C})$ with $W(A) \subset \Omega$,

$$K_A(t) = q_G(t)(G(t)I - A)^{-1} = \sum_{k \geq 0} b_k(A) t^{-k}, \quad E_A(t) = K_A(t) + K_A(t)^* \succeq 0 \quad (t \in \mathbb{T}). \quad (3.3)$$

Its coefficients decay geometrically and its series converges absolutely uniformly on $\mathbb{T}$. The scalar case applies to every fixed $z \in \Omega$.

Proof. Expand (3.1) at infinity after factoring at from the denominator. The constant term is one; the coefficient of t^{-k} is a polynomial of degree k with leading term $a^{-k}z^k$.

For the joint continuation, put

$$Q(w, t) = \frac{G(t) - G(w)}{t - w}.$$

Write $u = t^{-1}$, $v = w^{-1}$, and $G(t) = at + g(u)$, with g holomorphic for $|u| < r^{-1}$. Then

$$Q = a - uv \frac{g(u) - g(v)}{u - v}.$$

The divided difference is jointly holomorphic, with its derivative value on the diagonal. Off the finite diagonal, $Q \neq 0$ by univalence of G ; on that diagonal $Q = G'(t) \neq 0$; on either reciprocal axis, including their intersection, $Q = a \neq 0$. Consequently

$$B(w, t) = t \frac{\partial_t Q(w, t)}{Q(w, t)} = -u \frac{\partial_u Q}{Q}$$

is holomorphic on the reciprocal bidisk and vanishes on both axes. Its Taylor series has only the strictly mixed powers in (3.2). Cauchy estimates on the closed reciprocal bidisk of radius ρ^{-1} give $|\beta_{ij}| \leq C_\rho \rho^{i+j}$ for every $r < \rho < 1$. For a compact subset of the product collar, choose ρ above r but smaller than one and than the moduli of both coordinates on that compact set. The estimate then dominates the double series by a product of convergent geometric series. This proves normal convergence throughout the product collar, including at infinity, and absolute uniform convergence on $\mathbb{T}^2$. Comparing coefficients at $t = \infty$ in the defining identity for B , first with fixed finite w in the collar, gives the displayed boundary formula for $b_j \circ G$.

For distinct $w, t \in \mathbb{T}$, the circle identity $2 \operatorname{Re}(t/(t - w)) = 1$ gives

$$m(w, t) = 2 \operatorname{Re} \frac{q_G(t)}{G(t) - G(w)} \geq 0.$$

Indeed $q_G(t)$ is the outward normal, so convexity gives $\operatorname{Re}(\overline{q_G(t)}(G(t) - G(w))) \geq 0$. Continuity of the mixed-series expression supplies the same sign on the diagonal. Every term in B and its conjugate has nonzero frequency in each variable, so integration in either variable leaves exactly one.

For the matrix resolvent, every eigenvalue of A lies in $W(A)$ by testing a unit eigenvector. Thus $G(t)I - A$ is invertible on and outside the circle. Compactness of the circle permits a smaller exterior collar on which the resolvent remains holomorphic, including infinity where $K_A(\infty) = I$. Expanding for large $|t|$ gives the coefficients $b_k(A)$ by the same polynomial identities as (3.1). Cauchy estimates in the reciprocal disk give geometric decay and absolute uniform convergence on $\mathbb{T}$.

Finally put $R_t = G(t)I - A$. Direct congruence gives

$$R_t^*(K_A + K_A^*)R_t = q_G(t)R_t^* + \overline{q_G(t)}R_t.$$

At a unit vector x , its quadratic form is $2 \operatorname{Re}(\overline{q_G(t)}(G(t) - x^*Ax)) \geq 0$, by the supporting half-plane and $W(A) \subset \Omega$. Invertibility of R_t proves $E_A \succeq 0$. $\square$

These local double-layer calculations belong to the numerical-range method of Delyon–Delyon and Crouzeix–Palencia [12, 11]. The polynomials b_k are the classical Faber polynomials [5, 6].

Indeed, putting $\Phi = G^{-1}$ in the exterior, (3.2) gives $b_k(z) - \Phi(z)^k = O(z^{-1})$ at infinity; hence b_k is the polynomial part of Φ^k . The mixed coefficients β_{ij} are differentiated Grunsky coefficients, as the logarithmic derivative of Q in the proof shows. The classical coefficient normalization gives a weighted inequality [7, equations (1.1)–(1.3)]; the unweighted Hilbert–Schmidt contraction used here follows locally from convexity and the two angular marginals.

We next turn this kernel into a bounded Cauchy projection on boundary functions. For the density comparison we need estimates for its actual adjoint and a formula relating Laurent tests to the original-matrix field E_A .

For every integer $k \geq 1$, use

$$\mathcal{H}_k = L^2(\mathbb{T}, d\sigma; M_k(\mathbb{C})), \quad \langle u, v \rangle = \int_{\mathbb{T}} \text{tr}(u^* v) d\sigma, \quad \|u\|_2^2 = \langle u, u \rangle.$$

The pointwise norm is the unnormalized Hilbert–Schmidt norm. Let P_0, P_+, P_- be the orthogonal Fourier projections onto the constant, strictly positive, and strictly negative frequencies, respectively, and write $u = u_0 + u_+ + u_-$ with $u_0 = P_0 u$, $u_+ = P_+ u$, $u_- = P_- u$. Define

$$(Mu)(w) = \int_{\mathbb{T}} m(w, t) u(t) d\sigma(t).$$

Remark 3.2 (Correspondence with the direct proof). A reader who has already checked the exterior calculations in [18, Theorems 2.1 and 3.1] can compare the conventions by setting $h = G$, $\lambda = t$, $\mu = w$. Its correction satisfies $s(\lambda, \mu) = B(\mu, \lambda)$ and $s_{kj} = \beta_{jk}$. The resulting integral action is the same M , not its Hilbert adjoint. The local proof above is complete; readers familiar with that calculation can continue with the physical Cauchy interpretation below.

For continuous matrix-valued boundary data u , define the physical Cauchy integral

$$c_u(z) = \int_{\mathbb{T}} u(t) \frac{q_G(t)}{G(t) - z} d\sigma(t), \quad z \in \Omega.$$

For finite Laurent data, the scalar case of (3.3) allows termwise integration for every fixed $z \in \Omega$. If $u(t) = \sum_j U_j t^j$ is finite, it gives

$$c_u(z) = \sum_{j \geq 0} b_j(z) U_j. \quad (3.4)$$

Thus c_u is an entire polynomial, and the physical integral of t^j is b_j for $j \geq 0$, zero for $j < 0$. The boundary trace is taken *after* this interior identity is proved; no singular boundary substitution in the integral is needed.

Proposition 3.3 (The local projection theorem). *For finite Laurent data u , the polynomial c_u has boundary trace $\mathcal{C}u = c_u \circ G$. This map extends to a bounded projection on every $\mathcal{H}_k$. With $\tilde{\mathcal{C}}u = (\mathcal{C}(u^*))^*$, one has*

$$\mathcal{C} = P_0 + P_+ + MP_+, \quad \tilde{\mathcal{C}}u = (\mathcal{C}(u^*))^*.$$

Its actual Hilbert-space adjoints satisfy

$$\mathcal{C}^\dagger u = u_0 + u_+ + M^\dagger u_-, \quad \tilde{\mathcal{C}}^\dagger u = u_0 + u_- + M^\dagger u_+ = (\mathcal{C}^\dagger(u^*))^*. \quad (3.5)$$

All four operators preserve the mean, and

$$\|\mathcal{C}^\dagger u\|_2^2 + \|(\mathcal{C}^\dagger u)_0\|_2^2 \leq 2\|u\|_2^2 \quad (u \in \mathcal{H}_k). \quad (3.6)$$

If $F = F_\Omega \circ G$ with F_Ω holomorphic near $\bar{\Omega}$ and M_k -valued, then for every $u \in \mathcal{H}_k$,

$$\mathcal{C}(FCu) = FCu, \quad \mathcal{C}((Cu)F) = (Cu)F. \quad (3.7)$$

For every finite coefficient size and every matrix Laurent polynomial u ,

$$\int_{\mathbb{T}} E_A \otimes u \, d\sigma = c_u[A] + c_{u^*}[A]^*. \quad (3.8)$$

In particular, $\int E_A \, d\sigma = 2I_n$.

Proof. Positivity and the two marginal identities give the contraction locally: Jensen's inequality at each w and then integration imply

$$\|(\mathbf{M}u)(w)\|_{\text{HS}}^2 \leq \int_{\mathbb{T}} m(w, t) \|u(t)\|_{\text{HS}}^2 \, d\sigma(t), \quad \|\mathbf{M}u\|_2 \leq \|u\|_2.$$

The mixed expansion in (3.2) gives, for $j > 0$,

$$\mathbf{M}(t^j)(w) = \sum_{i \geq 1} \beta_{ij} w^{-i}, \quad \mathbf{M}(t^{-j})(w) = \sum_{i \geq 1} \overline{\beta_{ij}} w^i.$$

Both marginal identities say that $\mathbf{M}$ fixes constants and preserves means. The real kernel makes it commute with pointwise matrix adjoint. Its actual adjoint has kernel $m(t, w)$, and hence, for $i > 0$,

$$\mathbf{M}^\dagger(t^{-i})(w) = \sum_{j \geq 1} \overline{\beta_{ij}} w^j, \quad \mathbf{M}^\dagger(t^i)(w) = \sum_{j \geq 1} \beta_{ij} w^{-j}.$$

It is a contraction, fixes constants, preserves means, and commutes with pointwise adjoint. No symmetry of m is used. By boundedness and density of Fourier polynomials, the displayed actions extend to the closed strictly positive and strictly negative Fourier subspaces.

Equations (3.2) and (3.4) give $Cu = u_0 + u_+ + \mathbf{M}u_+$ for finite Laurent data. The three terms are orthogonal, so $\|Cu\|_2^2 \leq 2\|u\|_2^2$. Density defines the bounded extension, and the same formula gives $\mathcal{C}^2 = \mathcal{C}$. Taking its adjoint and using the strict Fourier exchange proves (3.5); conjugation by the antiunitary map $u \mapsto u^*$ gives the second formula. The weighted estimate follows from

$$\|\mathcal{C}^\dagger u\|_2^2 + \|(\mathcal{C}^\dagger u)_0\|_2^2 = 2\|u_0\|_2^2 + \|u_+ + \mathbf{M}^\dagger u_-\|_2^2 \leq 2\|u\|_2^2.$$

We need the physical interpretation also for analytic boundary traces. If $v = v_\Omega \circ G$ and v_Ω is holomorphic near $\bar{\Omega}$, then v is holomorphic on an annulus about $\mathbb{T}$. On a smaller closed annulus its Laurent series converges absolutely uniformly. If $r < |w| < 1$, then $G(w) \in \Omega$: an exterior or boundary value would also have a preimage with modulus at least one, contradicting univalence on the collar. For $|w| < 1$ sufficiently close to one, expansion of $t/(t-w) + B(w, t)$ in the Cauchy integral gives

$$c_v(G(w)) = \sum_{j \geq 0} v_j w^j + \sum_{i, j \geq 1} \beta_{ij} w^{-i} v_j.$$

The series converge uniformly after shrinking the annulus again, by the geometric Laurent bounds for v_j and the normally convergent mixed expansion. At the boundary the expression is $\mathcal{C}v$. Cauchy's formula gives $c_v = v_\Omega$ in Ω , and thus $\mathcal{C}v = v$ on $\mathbb{T}$. For Laurent data u , both $F_\Omega c_u$ and $c_u F_\Omega$ are holomorphic near $\bar{\Omega}$ by (3.4). Their traces are fixed by $\mathcal{C}$, giving both identities in (3.7) for finite Laurent u . Left and right multiplication by the bounded F are bounded on $\mathcal{H}_k$; Laurent approximation extends the two identities to arbitrary $u \in \mathcal{H}_k$.

Lastly, integrating (3.3) against a finite Laurent polynomial gives

$$\int_{\mathbb{T}} K_A \otimes u \, d\sigma = \sum_{j \geq 0} b_j(A) \otimes U_j = c_u[A]$$

by (3.4) and the prescribed base-first evaluation. Applying the same formula to u^* and taking the full matrix adjoint proves (3.8). The constant case gives mass $2I_n$. $\square$

In the density comparison below, we evaluate the physical Cauchy representative only for the Laurent test functions in (3.8). Arbitrary L^2 data in (3.7) are obtained by bounded extension and density. No continuation of a general $\mathcal{C}u$ through the boundary is assumed. Also, E_A is different from the normalized density Λ in Proposition 2.1: they have different masses and represent different expressions.

4 From extremal vectors to two positive density systems

We now return to the endpoint vectors constructed in Proposition 2.2. The boundary tools just proved provide two different positive systems: the mass-one density Λ for A' , and the mass-two resolvent field E_A for A . We compare their coefficient-space partial traces using the same vectors. Assume $\kappa > 1$ and write $R_S = S \otimes I_n$. Then

$$\|x\| = \|y\| = 1, \quad R_S x = \kappa x, \quad R_S y = y, \quad F_\Omega[A']x = y. \quad (4.1)$$

The partial-trace densities introduced below are n -by- n coefficient-space matrices; Λ and E_A remain base-space fields.

4.1 Equality in the common positive representation

Use the isometry $V = V_n$ of Proposition 2.1, and set

$$x_\Lambda = Vx = (\Lambda^{1/2} \otimes I_n)x, \quad y_\Lambda = Vy = (\Lambda^{1/2} \otimes I_n)y.$$

Let M_F multiply by $I_n \otimes F(t)$. The interpolation identity gives

$$1 = \|V^\dagger M_F Vx\| \leq \|M_F Vx\| \leq \|Vx\| = 1.$$

Equality in the first inequality puts $M_F Vx$ in the range of V : the orthogonal projection $VV^\dagger$ cannot preserve the norm of a vector with a nonzero perpendicular component. Thus $M_F Vx = Vy$. Equality in the second inequality says that the positive operator $I - M_F^\dagger M_F$ has zero quadratic form at Vx , so it annihilates Vx . Consequently, almost everywhere,

$$y_\Lambda = (I_n \otimes F)x_\Lambda, \quad x_\Lambda = (I_n \otimes F^*)y_\Lambda. \quad (4.2)$$

Take the partial trace over the base factor, meaning the sum of the diagonal base-space blocks, leaving a coefficient-space matrix:

$$p = \text{Tr}_{\text{base}}(x_\Lambda x_\Lambda^*), \quad q = \text{Tr}_{\text{base}}(y_\Lambda y_\Lambda^*), \quad r = \text{Tr}_{\text{base}}(y_\Lambda x_\Lambda^*). \quad (4.3)$$

They are bounded, hence in $\mathcal{H}_n$, and $p, q \succeq 0$. To see each multiplication order, write $x_\Lambda = \sum_a e_a \otimes \xi_a$ and $y_\Lambda = \sum_a e_a \otimes \eta_a$. Then $r = \sum_a \eta_a \xi_a^*$, while (4.2) gives $\eta_a = F\xi_a$ and $\xi_a = F^*\eta_a$. Therefore

$$p = F^* r, \quad r = F p, \quad q = r F^*. \quad (4.4)$$

The same convention identifies the mean without losing matrix information:

$$p = (X^* \Lambda X)^\top, \quad q = (Y^* \Lambda Y)^\top, \quad r = (X^* \Lambda Y)^\top. \quad (4.5)$$

Indeed, the (j, k) entry of the cross density is

$$\sum_a (\Lambda^{1/2} Y)_{aj} \overline{(\Lambda^{1/2} X)_{ak}} = (X^* \Lambda Y)_{kj}.$$

Integrating (4.5) gives

$$r_0 = (X^* Y)^\top = 0, \quad \text{tr}(p_0) = 1. \quad (4.6)$$

Thus the entire matrix mean vanishes; the scalar identity $\text{tr}(r_0) = 0$ would not provide the Fourier orthogonality used below.

For every $u \in \mathcal{H}_n$, the cross density has the testing orientation

$$\langle r, u \rangle = \int_{\mathbb{T}} y_\Lambda^* (I_n \otimes u) x_\Lambda d\sigma = y^* \left(\int_{\mathbb{T}} \Lambda \otimes u d\sigma \right) x. \quad (4.7)$$

This follows from $\text{tr}(r^* u) = \sum_a \eta_a^* u \xi_a$. The diagonal formulas use (x, x) for p and (y, y) for q .

4.2 The original-matrix positive block

Use $E_A \succeq 0$ from Proposition 3.1, and define

$$x_e = (E_A^{1/2} \otimes I_n)x, \quad y_e = (E_A^{1/2} \otimes I_n)y.$$

Define p_e, q_e, r_e by the same three partial traces as in (4.3), now using x_e, y_e . Writing $x_e = \sum_a e_a \otimes \alpha_a$ and $y_e = \sum_a e_a \otimes \beta_a$ yields

$$\Delta = \begin{pmatrix} p_e & r_e^* \\ r_e & q_e \end{pmatrix} = \sum_a \begin{pmatrix} \alpha_a \\ \beta_a \end{pmatrix} \begin{pmatrix} \alpha_a \\ \beta_a \end{pmatrix}^* \succeq 0. \quad (4.8)$$

This positivity uses $W(A) \subset \Omega$ for the original matrix. No numerical-range containment for $A' = SAS^{-1}$ is used.

We now identify the new densities with projected versions of the first ones. For $M_n(\mathbb{C})$ -valued v holomorphic near $\bar{\Omega}$, similarity and (4.1) give

$$\begin{aligned} v[A] &= R_S^{-1} v[A'] R_S, & v[A]^* &= R_S v[A']^* R_S^{-1}, \\ y^* v[A] x &= \kappa y^* v[A'] x, & y^* v[A]^* x &= \kappa^{-1} y^* v[A']^* x. \end{aligned} \quad (4.9)$$

For diagonal pairings, the factors cancel for both the analytic operator and its adjoint.

Take a matrix Laurent polynomial u . Its physical representative c_u is holomorphic near $\bar{\Omega}$, so (3.8), (4.9), and (4.7) give

$$\begin{aligned} \langle r_e, u \rangle &= y^* (c_u[A] + c_u^*[A]^*) x \\ &= \kappa \langle r, \mathcal{C}u \rangle + \kappa^{-1} \langle r, \tilde{\mathcal{C}}u \rangle \\ &= \langle \kappa \mathcal{C}^\dagger r + \kappa^{-1} \tilde{\mathcal{C}}^\dagger r, u \rangle. \end{aligned}$$

In the second term the boundary test is $(\mathcal{C}(u^*))^*$, which is $\tilde{\mathcal{C}}u$; it is not an independent analytic function evaluated at A . The corresponding diagonal computations have no similarity factor. Since Laurent polynomials are dense in $\mathcal{H}_n$, we obtain

$$\begin{aligned} p_e &= \mathcal{C}^\dagger p + \tilde{\mathcal{C}}^\dagger p, & q_e &= \mathcal{C}^\dagger q + \tilde{\mathcal{C}}^\dagger q, \\ r_e &= \kappa \mathcal{C}^\dagger r + \kappa^{-1} \tilde{\mathcal{C}}^\dagger r. \end{aligned} \quad (4.10)$$

All functions here lie in the same Hilbert–Schmidt L^2 space.

5 The ordered comparison and the constant two

The preceding construction supplies the hypotheses of the following comparison. We state it for arbitrary coefficient size to expose its dimension-free content. The comparison uses one nonzero field $g = \mathcal{C}^\dagger r$: the diagonal norms are at most $2\|g\|_2$, whereas the cross norm is at least $\kappa\|g\|_2$. Positivity of the block then forces $\kappa \leq 2$. The proof below keeps the left and right multiplication orders separate when obtaining the two diagonal bounds.

Lemma 5.1 (Ordered density comparison). *Let $k \geq 1$, and let $F = F_\Omega \circ G$ be M_k -valued, with F_Ω holomorphic near $\bar{\Omega}$ and $\|F(t)\| \leq 1$ on $\mathbb{T}$. Suppose $p, q, r \in \mathcal{H}_k$ satisfy*

$$p = p^*, \quad q = q^*, \quad p = F^* r, \quad q = r F^*, \quad r_0 = 0, \quad \text{tr}(p_0) > 0.$$

For a positive number κ , define

$$p_e = \mathcal{C}^\dagger p + \tilde{\mathcal{C}}^\dagger p, \quad q_e = \mathcal{C}^\dagger q + \tilde{\mathcal{C}}^\dagger q, \quad r_e = \kappa \mathcal{C}^\dagger r + \kappa^{-1} \tilde{\mathcal{C}}^\dagger r.$$

If the $2k$ -by- $2k$ block

$$\Delta(t) = \begin{pmatrix} p_e(t) & r_e(t)^* \\ r_e(t) & q_e(t) \end{pmatrix} \succeq 0$$

for almost every $t \in \mathbb{T}$, then $\kappa \leq 2$.

Proof. Put $g = \mathcal{C}^\dagger r$, $P_1 = \mathcal{C}^\dagger p$, $P_2 = \mathcal{C}^\dagger q$. On $\mathcal{H}_k$, the adjoint of left multiplication by F is left multiplication by F^* ; the adjoint of right multiplication is right multiplication by F^* , because

$$\text{tr}((uF)^* v) = \text{tr}(F^* u^* v) = \text{tr}(u^* v F^*).$$

Taking the Hilbert adjoints of both identities in (3.7) therefore gives, with the factors on their original sides,

$$P_1 = \mathcal{C}^\dagger (F^* g), \quad P_2 = \mathcal{C}^\dagger (g F^*). \quad (5.1)$$

If $g = 0$, the first identity makes $P_1 = 0$, contradicting mean preservation and $\text{tr}(p_0) > 0$. Thus $g \neq 0$.

Each P_j has only nonnegative Fourier powers, and its mean is Hermitian. Fourier orthogonality gives

$$\|P_j + P_j^*\|_2^2 = 4\|(P_j)_0\|_2^2 + 2\|(P_j)_+\|_2^2 = 2(\|P_j\|_2^2 + \|(P_j)_0\|_2^2).$$

Since $\|F\| \leq 1$, both $F^* g$ and $g F^*$ have Hilbert–Schmidt L^2 norm at most $\|g\|_2$. Apply (3.6) separately to the two inputs in (5.1). Using the star relation in (3.5) and $p = p^*$, $q = q^*$, we obtain

$$\|p_e\|_2^2 \leq 4\|g\|_2^2, \quad \|q_e\|_2^2 \leq 4\|g\|_2^2. \quad (5.2)$$

Put $h = \tilde{\mathcal{C}}^\dagger r$. Mean preservation and the whole matrix equality $r_0 = 0$ put g in the strictly positive Fourier subspace and h in the strictly negative one. Therefore

$$\|r_e\|_2^2 = \kappa^2 \|g\|_2^2 + \kappa^{-2} \|h\|_2^2 \geq \kappa^2 \|g\|_2^2. \quad (5.3)$$

Finally, set $J = \text{diag}(I_k, -I_k)$. Positivity of both Δ and $J\Delta J$ gives

$$0 \leq \text{tr}(\Delta^{1/2} (J\Delta J) \Delta^{1/2}) = \text{tr}(\Delta J\Delta J) = \|p_e\|_{\text{HS}}^2 + \|q_e\|_{\text{HS}}^2 - 2\|r_e\|_{\text{HS}}^2.$$

The entries are in L^2 , so this inequality is integrable. Combining it with (5.2) and (5.3) yields

$$2\kappa^2 \|g\|_2^2 \leq 2\|r_e\|_2^2 \leq \|p_e\|_2^2 + \|q_e\|_2^2 \leq 8\|g\|_2^2.$$

Since $g \neq 0$, division proves $\kappa \leq 2$. □

Completion of Theorems 1.2 and 1.3. If $\kappa = 1$, the required bound already holds. Otherwise Proposition 2.2 and Section 4 supply all the hypotheses of Lemma 5.1 at auxiliary size $k = n$: (4.4) gives both multiplication orders, (4.6) gives the zero matrix mean and positive trace, (4.8) supplies positivity, and (4.10) identifies its entries. Hence $\kappa \leq 2$. Proposition 2.1 already used this same metric and density for every independently chosen coefficient size m , which proves all statements of the two theorems. $\square$

6 From admissible domains to bounded Hilbert-space operators

We first pass from admissible enclosing domains to arbitrary finite-matrix numerical ranges. The Gaussian-distance construction below supplies the required domains, including when the numerical range is a point or a segment. A finite support-exponential construction gives an alternative [18, Section 4]. We then transfer the polynomial estimate by finite compression and pass to the rational and holomorphic functional calculi.

Corollary 6.1 (The sharp complete polynomial inequality for matrices). *For every pair of positive integers n, m , every $A \in M_n(\mathbb{C})$, every nonnegative integer d , and every choice of matrices $B_0, \dots, B_d \in M_m(\mathbb{C})$, the polynomial $P(z) = \sum_{j=0}^d B_j z^j$ satisfies*

$$\|P[A]\| = \left\| \sum_{j=0}^d A^j \otimes B_j \right\| \leq 2 \max_{z \in W(A)} \|P(z)\|.$$

The constant is independent of both matrix sizes and the degree. No smaller constant holds uniformly over these choices.

Lemma 6.2 (Gaussian analytic outer approximation). *For every nonempty compact convex $K \subset \mathbb{R}^2$ and every $\epsilon > 0$, there is an admissible Ω_ϵ such that*

$$K \subset \Omega_\epsilon, \quad \overline{\Omega}_\epsilon \subset \{x : \text{dist}(x, K) < \epsilon\}.$$

Proof. The distance function d_K is convex and 1-Lipschitz. For convexity, choose nearest points in K to two given points, use their convex combination as a competitor, and apply the triangle inequality. Let Z be a standard Gaussian in $\mathbb{R}^2$ and put

$$h_\delta(x) = \mathbb{E} d_K(x - \delta Z) + \delta |x|^2, \quad c = \mathbb{E}|Z|.$$

The Lipschitz bound gives the uniform estimate

$$|\mathbb{E} d_K(x - \delta Z) - d_K(x)| \leq c\delta.$$

The convolution is real analytic: write it against the translated Gaussian density, complexify both coordinates of x , and use Gaussian domination on every compact set of complex coordinates. The factor $d_K(y)$ has at most linear growth, so the integral defines a holomorphic function of those coordinates. Convexity passes through convolution; therefore h_δ is proper and strictly convex, with Hessian at least $2\delta I$.

Put $R_K = \max_{x \in K} |x|$, and choose $\delta > 0$ with $c\delta + \delta R_K^2 < \epsilon/2$ and $c\delta < \epsilon/2$. Then $h_\delta < \epsilon/2$ on K , while $h_\delta \leq \epsilon/2$ implies $d_K \leq h_\delta + c\delta < \epsilon$. Thus $\Omega_\epsilon = \{h_\delta < \epsilon/2\}$ has the desired containments and is bounded, open, and convex. The chosen level is above the minimum. A critical point of a differentiable convex function is a global minimum, so the gradient is nonzero on this level. The implicit function theorem gives a regular real-analytic boundary. From an interior point, each ray meets this compact boundary exactly once. Radial projection is a continuous bijection onto the circle and hence a homeomorphism. The boundary is therefore a Jordan curve. $\square$

Proof of Corollary 6.1. For completeness, the Toeplitz–Hausdorff convexity theorem [26, 13] follows here from two-dimensional compression. Compress to the span of unit vectors realizing any two chosen values. In dimension two, the rank-one orthogonal projections are

$$\frac{1}{2} \begin{pmatrix} 1+u & v+iw \\ v-iw & 1-u \end{pmatrix}, \quad u^2 + v^2 + w^2 = 1, \quad u, v, w \in \mathbb{R}.$$

Taking their trace against the compression is a real-affine map into $\mathbb{C} \simeq \mathbb{R}^2$. Its linear part has a nonzero kernel. Each point of the unit ball can be moved in a kernel direction to the sphere without changing its image, so the images of the sphere and ball coincide. That image is convex. The one-dimensional compression case is immediate. Compactness of $W(A)$ follows from compactness of the unit sphere.

Apply Lemma 6.2 to $K = W(A)$. For each fixed polynomial $P(z) = \sum_{j=0}^d B_j z^j$, with arbitrary $B_j \in M_m(\mathbb{C})$, Theorems 1.2 and 1.3 give

$$\left\| \sum_{j=0}^d A^j \otimes B_j \right\| \leq 2 \max_{\Omega_\epsilon} \|P(z)\|.$$

Uniform continuity on a fixed compact neighborhood of K makes the maxima on the right tend to $\max_K \|P\|$. This includes point and segment numerical ranges. It is a limit of scalar bounds for a fixed polynomial; no limiting similarity or boundary density on K is asserted. For sharpness take $n = 2$, $m = 1$, $P(z) = z$, and $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$. Its norm is two, whereas, for every unit $x \in \mathbb{C}^2$,

$$|x^* A x| = 2|x_1 x_2| \leq |x_1|^2 + |x_2|^2 = 1,$$

with equality when $x_1 = x_2 = 1/\sqrt{2}$. Hence $\max_{W(A)} |P| = 1$, proving optimality uniformly over all finite base sizes, coefficient sizes, and degrees. $\square$

The passage to arbitrary Hilbert spaces uses only finitely many iterates of the input to a polynomial. The closure of the numerical range is needed for the compact spectral-set formulation, because an infinite-dimensional numerical range need not be closed.

Corollary 6.3 (The sharp complete inequality for bounded operators). *Let $\mathcal{H}$ be a complex Hilbert space, with no separability assumption, and let $A \in \mathcal{B}(\mathcal{H})$. Suppose first that $\mathcal{H} \neq \{0\}$, and put*

$$W(A) = \{\langle x, Ax \rangle : x \in \mathcal{H}, \|x\| = 1\}, \quad K = \overline{W(A)}.$$

Then K is compact and convex and contains $\text{spec}(A)$. For every positive integer m , every nonnegative integer d , and every matrix polynomial $P(z) = \sum_{k=0}^d B_k z^k$ with $B_k \in M_m(\mathbb{C})$, the operator $P[A] = \sum_{k=0}^d A^k \otimes B_k$ on $\mathcal{H} \otimes \mathbb{C}^m$ satisfies

$$\|P[A]\| \leq 2 \sup_{z \in W(A)} \|P(z)\| = 2 \max_{z \in K} \|P(z)\|. \quad (6.1)$$

More generally, if U is an open neighborhood of K and $F = (f_{ij}) : U \rightarrow M_m(\mathbb{C})$ is holomorphic, its entrywise holomorphic evaluation, in the same base-first tensor convention, satisfies

$$F[A] := \sum_{i,j=1}^m f_{ij}(A) \otimes E_{ij}, \quad \|F[A]\| \leq 2 \max_{z \in K} \|F(z)\|. \quad (6.2)$$

Here E_{ij} are the standard matrix units and $f_{ij}(A)$ is given by the holomorphic functional calculus. In particular, K is a complete 2-spectral set for A : the same estimate holds at every finite coefficient size for every matrix-valued rational function whose poles lie outside K , with evaluation by the rational functional calculus. The constant two is optimal uniformly over all these choices.

If $\mathcal{H} = \{0\}$, every evaluation is the zero operator. With the supremum over the empty numerical range defined to be zero, the corresponding supremum-norm inequalities are $0 \leq 0$; the compact spectral-set assertion above is stated for nonzero $\mathcal{H}$.

Proof. Assume $\mathcal{H} \neq \{0\}$. We first prove the polynomial bound. With $e_1, \dots, e_m$ the standard basis of $\mathbb{C}^m$, every vector in the Hilbert tensor product has the exact form $\xi = \sum_{j=1}^m \xi_j \otimes e_j$, with $\xi_j \in \mathcal{H}$. For $\xi \neq 0$, define the nonzero finite-dimensional subspace

$$E = \text{span}\{A^k \xi_j : 0 \leq k \leq d, 1 \leq j \leq m\}, \quad A_E = Q_E A|_E,$$

where Q_E is the orthogonal projection onto E . Induction gives $A_E^k \xi_j = A^k \xi_j$ for $0 \leq k \leq d$: at each step the next iterate already belongs to E . Hence, as vectors in $E \otimes \mathbb{C}^m$,

$$P[A]\xi = \sum_{k=0}^d \sum_{j=1}^m A^k \xi_j \otimes B_k e_j = \sum_{k=0}^d \sum_{j=1}^m A_E^k \xi_j \otimes B_k e_j = P[A_E]\xi.$$

For each unit $x \in E$, orthogonality of Q_E gives $\langle x, A_E x \rangle = \langle x, Q_E A x \rangle = \langle x, A x \rangle$. With the conjugate-linear-first convention, these are exactly the numerical-range values used in Corollary 6.1. Thus $W(A_E) \subset W(A)$. Applying Corollary 6.1 in an orthonormal basis of E yields

$$\|P[A]\xi\| \leq 2 \max_{z \in W(A_E)} \|P(z)\| \|\xi\| \leq 2 \sup_{z \in W(A)} \|P(z)\| \|\xi\|.$$

Taking the supremum over unit ξ proves the inequality. This argument uses no sequence of subspaces exhausting $\mathcal{H}$.

The same two-vector compression argument used in the proof of Corollary 6.1 shows that $W(A)$ is convex. It is bounded by $\|A\|$, so K is compact and convex. Continuity of P and density of $W(A)$ in K give the equality of the supremum and maximum in (6.1). To check spectral containment, let $\lambda \notin K$ and $\delta = \text{dist}(\lambda, K) > 0$. For every $x \in \mathcal{H}$, Cauchy–Schwarz gives

$$\|(\lambda I - A)x\| \|x\| \geq |\langle x, (\lambda I - A)x \rangle| \geq \delta \|x\|^2.$$

Since $W(A^*) = \{\bar{z} : z \in W(A)\}$, the same argument gives $\|(\bar{\lambda} I - A^*)y\| \geq \delta \|y\|$ for every y . The first lower bound makes $\lambda I - A$ injective with closed range, and the second makes it dense because its orthogonal complement is the kernel of $\bar{\lambda} I - A^*$. Thus $\lambda I - A$ is invertible, proving $\text{spec}(A) \subset K$.

Now fix U and F as in the statement. Choose $\epsilon > 0$ such that the compact convex neighborhood

$$K_\epsilon = \{z \in \mathbb{C} : \text{dist}(z, K) \leq \epsilon\}$$

is contained in U . Its complement is connected. Runge’s polynomial approximation theorem, obtained by rational approximation and movement of poles to infinity in the connected complement [23, pp. 235–238], gives matrix polynomials P_ℓ with

$$\max_{z \in K_\epsilon} \|P_\ell(z) - F(z)\| \longrightarrow 0.$$

Indeed, apply the scalar theorem to each of the finitely many entries; entrywise uniform convergence implies convergence in matrix operator norm. By Lemma 6.2, choose an admissible Ω with $K \subset \Omega$ and $\overline{\Omega} \subset K_\epsilon$, and let $\Gamma = \partial\Omega$ have positive orientation. The holomorphic functional calculus, entry by entry, reads

$$F[A] = \frac{1}{2\pi i} \int_{\Gamma} (zI - A)^{-1} \otimes F(z) dz.$$

The integral converges in operator norm. For a polynomial, Cauchy's theorem moves the contour to a large circle, where the resolvent's Neumann expansion gives the prescribed base-first evaluation. Consequently

$$\|F[A] - P_\ell[A]\| \leq \frac{\text{length}(\Gamma)}{2\pi} \max_{z \in \Gamma} \|(zI - A)^{-1}\| \max_{z \in \Gamma} \|F(z) - P_\ell(z)\| \rightarrow 0.$$

The maxima of $\|P_\ell\|$ on K also converge to that of $\|F\|$. Applying (6.1) to P_ℓ and taking the limit proves (6.2). The larger compact neighborhood is used to obtain convergence on the fixed contour, not just on K .

A matrix-valued rational function whose poles lie outside K is holomorphic on a neighborhood U of K that avoids all its poles; the preceding choice then puts every pole outside $\overline{\Omega}$. Spectral containment makes its rational evaluation well-defined, and this agrees with the displayed contour evaluation. For example, the resolvent identity identifies the contour evaluation of $(z - a)^{-1}$ with $(A - aI)^{-1}$ when $a \notin \overline{\Omega}$; differentiation in a gives the higher pole orders, and partial fractions give every rational function. This proves the complete spectral-set assertion. Finally, the 2-by-2 scalar-coefficient example in the proof of Corollary 6.1 is already a bounded Hilbert-space operator and is both rational and holomorphic. It proves the stated uniform sharpness. $\square$

A Local conformal reconstruction

We give the classical extremal proof of Riemann mapping and the harmonic-reflection argument [2, Chapter 6, Section 1.1; Chapter 4, Section 6.5], then prove the collar patching and global univalence needed in Lemma 1.1. The extension argument is stated for a general analytic Jordan domain, since the inversion used for the exterior coordinate need not preserve convexity. For general background on boundary behavior of conformal maps, see [20].

Proof of Lemma 1.1. A disk coordinate for a bounded simply connected domain. Let $U \subset \mathbb{C}$ be a bounded simply connected domain and fix $z_* \in U$. Consider the injective holomorphic maps $\phi : U \rightarrow \mathbb{D}$ satisfying $\phi(z_*) = 0$. This family is nonempty, since a sufficiently small multiple of $z - z_*$ belongs to it. The supremum α of $|\phi'(z_*)|$ over the family is positive and finite: a disk centered at z_* with closure in U gives a uniform upper bound by Cauchy's derivative estimate.

Choose a sequence approaching this supremum. Cauchy's estimates and compact exhaustion give a subsequence converging locally uniformly to a holomorphic function ϕ . Its derivative at z_* has modulus α , so it is nonconstant; the maximum principle then gives $\phi(U) \subset \mathbb{D}$. The limit is injective. Indeed, if two distinct points had the same image, choose disjoint small closed disks about them on whose boundaries the limit minus this common image does not vanish. The argument principle preserves at least one zero in each disk for every sufficiently late member of the sequence, contradicting its injectivity.

Suppose that $w \in \mathbb{D} \setminus \phi(U)$. Necessarily $w \neq 0$. The function

$$q(z) = \frac{\phi(z) - w}{1 - \overline{w}\phi(z)}$$

is holomorphic and nonvanishing on the simply connected domain U , and therefore has a holomorphic square root h . It satisfies $|h| < 1$ and is injective, since $h^2 = q$ is injective. Set $\beta = h(z_*)$, so that $|\beta| = \sqrt{|w|}$. The injective disk-valued function

$$\psi(z) = \frac{h(z) - \beta}{1 - \overline{\beta}h(z)}$$

vanishes at z_* and has derivative magnitude

$$|\psi'(z_*)| = \frac{1 - |w|^2}{2\sqrt{|w|}(1 - |w|)} |\phi'(z_*)| = \frac{1 + |w|}{2\sqrt{|w|}} \alpha > \alpha.$$

This is impossible. Thus ϕ maps U conformally onto $\mathbb{D}$.

Reflection at a regular analytic boundary. Suppose now that ∂U is a regular real-analytic Jordan curve. For each $0 < r < 1$, the set $\phi^{-1}(r\mathbb{D})$ is a compact subset of U , because the conformal inverse is continuous on $\mathbb{D}$. Consequently $|\phi(z)| \rightarrow 1$ as z approaches ∂U from inside.

Fix $p \in \partial U$. A regular real-analytic parametrization of the boundary near p extends holomorphically with nonzero derivative. Its local holomorphic inverse straightens the boundary. After shrinking and choosing its sign, we therefore have a holomorphic coordinate $z = \chi(\xi)$ taking 0 to p , a real diameter to the boundary arc, and the upper half of a small disk to the part of U in this coordinate neighborhood. Choose the neighborhood disjoint from z_* . Then

$$u(\xi) = \log |\phi(\chi(\xi))|$$

is harmonic and strictly negative on the upper half disk, and is continuous with value zero on its real diameter.

Here the harmonic reflection can be obtained directly. On the boundary of a smaller full disk, prescribe u on the upper semicircle and the negative of its reflected values on the lower semicircle. These boundary values are continuous, including at the endpoints of the diameter. Their Poisson extension is odd under reflection in the real axis, hence vanishes on the diameter. Uniqueness for the Dirichlet problem on the upper half disk shows that it agrees there with u . Denote the resulting harmonic extension by $\hat{u}$.

The gradient of $\hat{u}$ at zero does not vanish. To see this, its first nonzero homogeneous Taylor term has the form

$$c \operatorname{Im}(\xi^k), \quad c \in \mathbb{R} \setminus \{0\}, \quad k \geq 1,$$

because $\hat{u}$ is harmonic and vanishes on the real diameter. Such a term exists since $u < 0$ on the upper half disk. If $k \geq 2$, $\sin(k\theta)$ takes both signs for $0 < \theta < \pi$; along two corresponding rays the leading term would force u to take both signs near zero. Thus $k = 1$.

Take a harmonic conjugate $\hat{v}$ on the full disk. The function $\exp(\hat{u} + i\hat{v})$ is holomorphic and nonvanishing, with nonzero derivative at zero. On the upper half disk it has the same modulus as $\phi \circ \chi$, so their quotient is a constant of modulus one. Adjusting that constant and returning to the physical coordinate extends ϕ holomorphically across p , with nonzero derivative there.

Patching and univalence on a neighborhood of the closure. We spell out the passage from these local extensions to one extension, including overlaps that may lie outside U . For each boundary point p , choose an extension disk $B(p, R_p)$ contained in its reflected coordinate neighborhood. Its extension agrees with ϕ throughout $U \cap B(p, R_p)$; shrink R_p so that its derivative is nonzero on the disk. Select finitely many points p_i such that the smaller disks

$$V_i = B(p_i, R_i/3), \quad R_i = R_{p_i},$$

cover ∂U , and let ϕ_i denote the extension on $U_i = B(p_i, R_i)$. If $V_i \cap V_j \neq \emptyset$ and $R_i \geq R_j$, then

$$|p_i - p_j| < \frac{R_i + R_j}{3} \leq \frac{2R_i}{3} < R_i.$$

Thus the connected open lens $U_i \cap U_j$ contains the boundary point p_j and meets U in a nonempty open set. Both extensions agree with ϕ there, so the identity theorem makes them equal on the entire lens. Their restrictions therefore patch with ϕ to a holomorphic function on

$$U \cup \bigcup_i V_i,$$

an open neighborhood of $\bar{U}$. We continue to call it ϕ .

The extended map takes ∂U into $\mathbb{T}$. Its nonzero derivative makes the restriction to the boundary a local diffeomorphism onto an arc of $\mathbb{T}$. The boundary image is consequently open in $\mathbb{T}$, and it is also nonempty and compact, so it is all of $\mathbb{T}$. There cannot be two distinct boundary points with the same image $\eta \in \mathbb{T}$. Indeed, choose disjoint local inverse neighborhoods about them. Each inverse, restricted to a sufficiently small neighborhood of η , takes its disk side into U : the boundary arc is mapped to $\mathbb{T}$, and the interior side is mapped into $\mathbb{D}$. Points in $\mathbb{D}$ sufficiently close to η would then have two preimages in U , contradicting interior injectivity. Since interior and boundary images lie in $\mathbb{D}$ and $\mathbb{T}$, respectively, ϕ is injective on $\bar{U}$.

It is in fact injective on an open neighborhood of $\bar{U}$. Otherwise there would be distinct pairs z_j, w_j , approaching the compact set $\bar{U}$, with $\phi(z_j) = \phi(w_j)$. Subsequence limits $z, w \in \bar{U}$ have the same image, hence $z = w$. Both sequences would eventually lie in one local inverse neighborhood of z , a contradiction. The image of a sufficiently small such neighborhood is open and contains $\bar{\mathbb{D}}$. Its holomorphic inverse gives a univalent extension of ϕ^{-1} to a neighborhood of $\bar{\mathbb{D}}$.

Applying these conclusions to the admissible domain Ω gives the interior coordinate f and both of its asserted extensions.

The exterior coordinate. Fix $z_0 \in \Omega$ and apply the inversion

$$\iota(z) = \frac{1}{z - z_0}, \quad \iota(\infty) = 0,$$

to the exterior of $\bar{\Omega}$, including infinity. Its image U_{ext} is a bounded domain containing zero. More explicitly, convexity and $z_0 \in \Omega$ give a finite positive boundary radius $R(\theta)$ on each ray $z_0 + re^{i\theta}$, and

$$U_{\text{ext}} = \{0\} \cup \{se^{-i\theta} : 0 < s < R(\theta)^{-1}\}.$$

This domain is star-shaped about zero, hence simply connected. Boundedness follows also from a positive disk about z_0 contained in Ω ; a neighborhood of zero is in the image because Ω is bounded. Inversion preserves the regular real-analytic Jordan boundary, since its pole z_0 is not on that boundary.

By the preceding arguments there is a conformal map $\phi_{\text{ext}} : U_{\text{ext}} \rightarrow \mathbb{D}$ with $\phi_{\text{ext}}(0) = 0$, whose inverse h extends univalently to a disk $\{\zeta : |\zeta| < \rho\}$ for some $\rho > 1$. Here $h(0) = 0$ is its only zero on this larger disk, and it is simple. Define

$$G(t) = z_0 + \frac{1}{h(1/t)}, \quad |t| > r_0 := \rho^{-1}.$$

The function G is univalent on this region. It maps $|t| > 1$ onto the exterior of $\bar{\Omega}$, takes infinity to infinity, and has a simple pole there. Its convergent Laurent expansion is therefore

$$G(t) = at + b + \sum_{\ell \geq 1} c_\ell t^{-\ell}, \quad a = h'(0)^{-1} \neq 0.$$

In particular its restriction to $\mathbb{T}$ is a smooth diffeomorphism onto $\partial\Omega$.

Orientation and normal direction. For $t = e^{i\theta}$, the radial and angular derivatives are

$$\left. \frac{\partial}{\partial r} G(re^{i\theta}) \right|_{r=1} = tG'(t), \quad \frac{d}{d\theta} G(e^{i\theta}) = itG'(t).$$

They are nonzero and perpendicular. Increasing r enters the exterior of $\overline{\Omega}$, so $tG'(t)$ points in the outward normal direction. The tangent $itG'(t)$ is consequently the positive boundary tangent. The interior conformal map f preserves boundary orientation: locally it maps the interior side of $\partial\Omega$ to the interior side of $\mathbb{T}$, and its derivative preserves orientation in the plane. Thus $f \circ G : \mathbb{T} \rightarrow \mathbb{T}$ is a smooth orientation-preserving diffeomorphism. This proves every assertion of the lemma. $\square$

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